

Gendered Mobility: Investigating how gender influences the Physical Mobility of Four Street Reclaimers in Johannesburg



BA Honours Research Report

Name: Amogelang Mokgadi Mokobane

Student No: 859114

Supervisor: Dr Melanie Samson

University of the Witwatersrand: School of Geography,
Archaeology and Environmental Studies

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Declaration

I, Amogelang Mokgadi Mokobane, declare that this report for a BA Geography (honours) degree at the University of Witwatersrand in the School of Geography, Archaeology and Environmental Studies, is my own work. I hereby declare that this honours report has not been submitted at this or any other institution at any time. All sources utilised in this report have been acknowledged by means of referencing.

Abstract

The high levels of unemployment in South Africa have excluded numerous people from the formal economy and forced them to venture into the urban informal economy. The recycling of waste has become a viable economic endeavour for the urban poor. Street reclaimers recover materials from households, businesses and government buildings. The work that street reclaimers do requires a lot of movement through the streets. The easier it is to travel through the streets and the more access that reclaimers have to places with materials, the more money they can make. Physical mobility, which is the ability to move freely and easily through space, determines the success of the reclaimer. In that regard, there are various factors that affect the reclaimer's ability to move easily through space. This study is concerned about factors related to gender that affect the physical mobility of male and female reclaimers. Two male and two female reclaimers were interviewed and observations were carried out to obtain a better understanding of how mobility and space are gendered. Furthermore, the movement of street reclaimers was recorded in an effort to deduce whether women move differently to men and if they do, what are the factors that influence the difference in movement. The present study found that in the context of the work that street reclaimers do and their everyday experiences, the space in which they move is gendered and their gender negatively and sometimes positively influences their physical mobility.

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1. Introduction

The high levels of unemployment in South Africa have excluded numerous people from the formal economy and forced them to venture into the urban informal economy (Schenck and Blaauw, 2011). The recycling of waste has become a viable economic endeavour for the urban poor. Street reclaimers have taken advantage of this niche in the informal market by recovering reusable or recyclable materials thrown away by others. These salvaged materials are primarily obtained from the streets but particularly around households and businesses then sold to buy-back centres (Schenck and Blaauw, 2011).

Indeed, the recycling of waste has opened doors for the urban poor to gain an income but women are not able to fully exploit the market due to various reasons such as violence, prejudice, gender roles and insecurity in the urban city (Dias, 2015). All of these factors affect a female recycler's ability to move through the urban city. Considering that walking through the urban city is an important aspect of street reclaiming, it is useful to investigate the factors that limit or encourage the physical mobility of female street reclaimers.

Furthermore, it is equally imperative to investigate the factors that influence the physical mobility of male reclaimers in order to conduct a comparative analysis of the two. From the early 1980s, feminist geographers and planners began examining the various ways in which women and men experience space. In particular, they analysed the spatial expectations about women and men as well as their ability to move through urban spaces. They argued that men and women move around urban spaces differently

because their perceptions of space and their daily experiences influence their daily walking practices and sense of belonging (Manolache, 2013). In that regard, mobility becomes gendered. The ability and inability to move around the urban city is essentially influenced by varying elements that are unique to women and men respectively.

The present study, within the context of Johannesburg, will investigate how a street reclaimer's gender influences their physical mobility in the city. It aims to do so by interviewing two male and two female street reclaimers and mapping their movement around the city. This will be done in an effort to explore how factors that cause gendered mobility have the potential to limit or increase the access that men and women have to the informal sector and in this case, street reclaiming.

1.1 Rationale

Scholars affirm that there is a need for more participatory mapping that reveals gendered space, mobility and resources because these factors have the ability to marginalise women and men alike. Almahmood *et al.* (2017) and Rocheleau *et al.* (1995) support this view by maintaining that there are numerous works of literature that acknowledge that the physical characteristics of some urban cities restrict or encourage the mobility of women and men differently but there is room for more material on the cultural, religious and social factors that restrict or encourage mobility.

My study will consider both the physical and non-physical aspects that restrict or encourage physical mobility in an effort add a small contribution to the already existing dialogue regarding this subject. I seek to contribute the perspective of gendered space and gendered mobility by mapping the movement of male and female waste pickers

who are poor and black. Participatory maps that illustrate gendered mobility or gendered space in urban areas have been produced by scholars such as Wiesner (2015), Almahmood *et al.* (2017), Ranade (2007) and Rocheleau *et al.* (1995) but these maps have dealt with ecological resources, physical aspects of the city or they did not include the South African context. While some authors such as Lee *et al.* (2015) have mapped out the movement of waste pickers, they have not mapped out their gendered movement.

I believe that this study is able to highlight the factors that hinder or encourage the mobility of male and female waste pickers in Johannesburg and is useful in understanding how space in the areas that street reclaimers move is gendered. Furthermore, it demonstrates how physical mobility is gendered.

1.2 Aim

The main aim of this report is to use the data that I acquire from the four street reclaimers that I worked with to gain a better understanding of how space and mobility in Johannesburg is gendered. Understanding the different factors involved in gendering space and mobility will help me to confidently conclude whether female reclaimers are able to move freely and easily in Johannesburg and fully exploit the recycling market for a decent income.

1.3 Research Question and Objectives

This research report seeks to provide answers to the following question: How does a street reclaimer's gender influence their physical mobility in the city? Moreover it seeks to achieve the following objectives:

1. Find out how male and female street reclaimers navigate the streets of Johannesburg (mode of transport).
2. Uncover the rationale behind how they decide which routes to take and where to go.
3. Observe and Map the routes that street reclaimers take.
4. Investigate and analyze key elements that street reclaimers consider before they navigate through space.
5. Discover some of the factors that negatively and positively affect the physical mobility of street reclaimers.
6. Gain insight into how reclaimers think gender affects physical mobility.

1.4 Overview of Report

The first chapter of this report located above laid out a brief introduction, rationale and aim of this research report in an effort to provide a general understanding of the ideas that will be dealt with in the rest of the text. The second chapter comprises of a literature review that will provide an engagement with what other people have said on key themes that relate to this study. The third chapter will outline the methodology and methods that I used to collect my data and further achieve my objectives. The fourth chapter will include the results obtained from data collected and a discussion of said

results. The fifth and final chapter will contain the concluding remarks of this report and possibilities for further research.

2. Literature Review

This literature review will discuss street reclaimers, gendered space, gendered mobility, participatory mapping, intersectionality and articulation. These concepts are all instrumental in understanding the participants involved in this study as well as the factors that play a role in engendering gendered mobility and space.

2.1 Defining Street Reclaimers

According to Schenck and Blaauw (2011), up to 2% of the world's population is involved in recycling or reclaiming materials for resale. It has been affirmed that people who venture into reclaiming are either unemployable in the formal economy or they are unable to find jobs in the formal economy. Medina (1997), established that people recycle waste or used material due to poverty. In a country plagued by unemployment, street reclaimers (also known as street waste pickers) have found a way to earn a living by becoming entrepreneurs who have identified materials that are in demand, areas where that material can be found and buy-back centres that will purchase it.

In that regard, this research report defines street reclaimers as individuals whose livelihoods depend on the collection and sale of recyclable materials (Langenhoven and Dyssel, 2007). They are called street reclaimers because they walk around in the streets and reclaim materials such as plastic, paper and glass that have been discarded by people.

2.2 Gendered Space

This study will consider space in terms of social relations and physical features such as urban infrastructure. There are numerous definitions of space in academia. Foucault thought of space as a stationary, cold and lifeless entity (Massey, 1994). Conversely, Massey (1994) believes that space is not neutral and immobile but it is a dynamic entity that is constantly changing with time even when it is not occupied. I agree with Massey's view of space but I appreciate Gupta and Ferguson's (1992) viewpoints on space as a physical and non-physical phenomenon which originates as space but becomes 'place' as we get to know it better and attach meaning to it. They maintain that people become aware of the freedom and threat of space once they have places where they feel safe and secure. For example, when one experiences the warmth and security of one's home, the city streets may seem dangerous or hostile.

Foucault's definition of space is one-dimensional and does not consider the living entities that exist in space and its fluidity. While Massey's conception of space is valid, Gupta and Ferguson's definition rings true because it is more detailed and expressive.

Now that the concept of space has been explained, *public space* needs to be defined because this is the sphere in which the street reclaimers conduct their business, in the public space of the urban city. In his book *Social Justice and the Fight for the Public Space*, Mitchell (2003) defines public space as "sites of interaction, encounter and the support of strangers" (p.3). The definition goes on to mention that the public space is a place that people share and enjoy unrestricted access to. Fenster (2005), disagrees with this definition because he argues that the 'public space' is not public because the word 'public' suggest openness, freedom and a sense of community. He believes that people such as women and minorities are restricted from having free unrestricted

access to the so-called public space. He states that some feminists believe that the public space is a Caucasian middle-upper class, heterosexual male domain.

It is fair to state that some urban cities are white, male-dominated arenas, some urban city centres such as central Johannesburg are dominated by black men. But the fact still remains that the black and white male-domination of public spaces still makes it difficult for all women to walk in the streets, especially alone. Kinnvall (2016), reiterates this point by highlighting that daily traumas women experience everyday play a vital role in creating notions of gendered space in urban cities. Everyday trauma can be any unpleasant experience that they encounter in cities. Trauma causes them to feel insecure and this insecurity influences the way in which they move around the city.

Fear of crime and harassment restrict women's movement in public spaces. This fear has the potential to limit their opportunities, independence, and sense of belonging and enjoyment of public spaces (Day, 2001). While it is possible that the perception of danger may influence women to adhere to social norms that paint them as vulnerable and defenceless in public spaces, it cannot be denied that women experience crime in a real way. A study by Day (2001), revealed that men who reside in Irving, California agreed that the area is completely safe but when asked if it is safe for women they affirmed that it is in fact unsafe for women though only a few had actually heard of or seen any real crime occur. The fear of crime and victimization that a woman feels in public spaces is sometimes fuelled by her lived experiences or the experiences of others. However, it has been proven that most incidents of sexual assault and victimization against women are not carried out by strangers in public spaces but in private spaces by people they know.

The image of women as fragile and vulnerable creatures has been painted by men in order to construct and maintain their masculine gender identities (Day, 2001). Women are perceived as endangered in public spaces as opposed to men who are strong and can defend themselves. Within the context of this perception, men are able to affirm their masculine claim to public space. Perceived threat of crime, lived experiences and perceptions of masculine as well as feminine gender identities all contribute to gendering space and turning public space into a man's arena.

2.2 Gendered Mobility

Physical mobility is a term that used to describe the ability of different people to navigate a space (Institution for Justice and Reconciliation, 2016). Furthermore, people move around in the environment in order to accomplish goals. A range of factors can have an effect on how people are able to achieve their goals due to their ability to move around freely and easily. These include, how one can or is unable to travel to different places in pursuing of and achievement of their goals (Institution for Justice and Reconciliation, 2016). Scholars in the field of geography have for a long time been unable to engage deeply and understand the relationship between mobility and gender (Catta, 2008). It can be argued that women experience the world differently compared to men. Mobility becomes gendered because of the different ways in which women and men navigate different spaces. For example, women's mobility may be constrained due to their inability to move into certain spaces because of perceptions of safety and biological reasons related to sanitation (Cattan, 2008).

2.3 Participatory Mapping

Women and men's narratives reveal the connection between their daily walking practices and their sense of belonging in the urban city (Fenster, 2005). Their perceptions of space such as believing that they do not belong in certain areas due to their social class, perceived threats to security, daily experiences and gender roles influences their ability to move through urban spaces. In their paper Almahmood *et al.* (2017), used GPS data and other data they collected to map out the movement of the young adults in order to prove that men but especially women's movement around the city is restricted by gender norms and traditions. They found that women frequent areas that justify their presence in the city such as shopping malls while men are free to frequent all areas of the city. Their article reiterates why participatory mapping is such an important tool that can help to reveal gendered space and the factors that cause it. Female waste pickers in Cape Town expressed their fear of rape, theft, kidnapping and stabbing when they navigate the streets for recyclable and reusable materials (Benson and Nandi Vanqa-Mgijima, 2009).

A number of females have been abducted during this year in Braamfontein. On the fifteenth of May this year, a female Wits student was abducted on the Mandela Bridge. This encouraged an "All Men" Residence at Wits to create a buddy system where the men would escort the women home to their respective residences so that they can evade attacks (Mmotla, 2017). The reality of the urban city is that women are not welcome. They do not enjoy free, unrestricted access because they are constantly being made to feel insecure. There is no freedom in fear. Urban streets are male

spaces because women cannot access some of them due to feelings of insecurity, time of day, access to facilities, cultural values and morals. Most of the texts I read about gendered participatory mapping stress the need for more participatory mapping by communities that illustrates gendered movement and space because these maps have the ability to visually depict the social and economic injustices that women face in the urban city (Rocheleau *et al.* 1995). Almahmood *et al.* (2017) sustain that they decided to map the movement of women and men in urban city because there is a minimal amount of literature that considers the social, cultural and religious factors that influence the way in which men and women in Saudi Arabia navigate the city.

The authors believe that they would be offering new perspectives that is different from literature that only highlights the spatial aspects of the city which hinder the movement of women such as distances, availability of public toilets and places that are closed off and badly lit. They conclude the article by mentioning that women walk in 'privatized' public spaces such as shopping malls. This supports Fenster's (2005), view that women are not welcome in public spaces so they stay in private spaces such as the home.

Almahmood *et al.* (2017) and Fenster's (2005), findings also reiterate Gupta and Ferguson's (1992), belief that people view space as a dangerous and hostile arena when they have places that they attach meaning to. Shopping malls are places where women can go without being frowned upon for walking around and homes are safe places where the woman feels secure. While Rocheleau *et al.* (1995) agree with Almahmood *et al.* (2017), in terms of having more maps that illustrate the different experiences of men and women, they are not concerned with the religious, cultural and social factors that influence movement. They are concerned with the equal distribution and ownership of resources and to show the knowledge that women have about ecological conservation.

2.4 Intersectionality

Intersectionality is used to describe how social categories such as race, ethnicity, class, gender and sexuality become naturally intertwined in addressing a certain individual or a group of people. This creates an overlap of systems of discrimination, oppression and disadvantages (McCall, 2005). According to Yuval-Davis (2006), she argues that intersectionality originated from a feminist politics in terms of able unable to resolve the issue of Black feminists from the White feminists. This meant that Black feminists could not separate themselves as being women without their race, which is that they are Black.

It is integral to discuss the intersections between gender, race and class because they determine how street reclaimers embody experiences of everyday life. Reclaimers are predominantly black and poor so it useful to consider these intersections because their and race and class are institutions which are oppressive. Female reclaimers are the most disadvantaged by these social constructions because they experience unjust treatment due to their gender.

2.5 Articulation

Indeed, streets are socially conceived as men's walkscapes but we need to consider the types of men who dominate Johannesburg City. This is where articulations of race,

gender and class come in. Articulation is defined as the complex intersections between social constructs such as race, gender and class. These social constructs are formed together and do not exist independent to each other (Hall, 1996).

Walkscapes of the inner city are dominated by black people who use walking as a mode of transport, black hawkers, black beggars, black homeless people and black street performers, among others. The streets of the inner city are dominated by people who may be below middle class, poor or involved in the informal economy. Street reclaimers who frequent these walkscapes are also dominantly black and below middle-class. Street reclaimers are predominantly black, male and below middle class but there are coloured South Africans, some white South Africans who are involved in reclaiming. All of them are generally very poor (Benson and Nandi Vanqa-Mgijima, 2009 *et al*). There are not many female reclaimers because of insecurity, ability to carry waste, distance needed to travel and gender stereotypes. But, female waste pickers that are there are black and below middle-class. Walkscapes in the inner city are dominated by black people who may be below middle class, poor or involved in the informal economy. Particularly one is likely to see black men. Street reclaimers exist in these walkscapes therefore, it is important to discuss these articulations because they provide a profile of the average type of street reclaimer one is likely to find in the inner city. More importantly, they encourage one to consider why women are not present in these walkscapes.

In conclusion, space is gendered because women and men attach meaning to it based on their culture, religions and lived experiences. These elements affect the ways in which men and women perceive and experience space. These perceptions of space and other factors related to their gender, affects their physical mobility, in that it affects their ability to move freely and easily in their environment to achieve their goals.

Mapping the different ways in which men and women move in urban spaces and the areas they frequent can highlight the factors that engender gendered mobility and gendered space. Maps and discussions on gendered mobility and space have revealed that women are marginalised in space and their physical mobility is impaired due to factors that place them in positions that are subordinate to men. Different genders do perceive and experience space differently but other social constructs such as race and class further influence people's interaction with space. For example, black women experience space differently to white women. Considering that race and class are socially constructed to the detriment of black and poor people, it is important to understand the lived experiences of street reclaimers in that context.

This review of existing literature regarding how mobility and space are gendered is instrumental in revealing that the research question can be answered by studying the movement of street reclaimers and gaining insight into their lived experiences. The existing literature contains an understanding of gendered mobility and space and has revealed that these concepts are present in urban spaces. The information obtained from the literature helped me identify how I can add a minor perspective of gendered mobility and space to further support the existing works.

3. Research Methodology

Identifying the right methodology (how and why knowledge is obtained) and methods (tools, techniques or strategies used to obtain data) is imperative because it ensures that the correct data that will answer the research question will be obtained (Clough and Nutbrown, 2012). Without the correct methodology and methods, one would collect data that would be of no use because it would not aid in answering the research question. This chapter will outline the study area, methodology and methods that were utilized and justify my choices.

3.1 Study Area

The study was carried out in Brixton, Linden, Mayfair, Westdene, Fairland and Emmarentia which are situated in Johannesburg, South Africa.

I wanted to conduct my study in Johannesburg city because I can easily access places in the area. My home and varsity is in Johannesburg city and I know most areas well. I also chose it because there are numerous street reclaimers in the area due to the generous number of buy-back centres, businesses and suburbs where they can obtain materials.

I wanted to conduct my research in Johannesburg but the decision to conduct it in the specific areas mentioned was made by the individual participants. I chose the days that I would travel with them and they told me where they collect materials on those specific days. All four of the participants that took part in this study live in Bekezela, Newtown. The areas where they collect materials are usually not more than 30km from Bekezela.

I devoted a fair amount of time throughout the year to spending as much time in Bekezela as I could before I conducted my interviews, observations and field research, I spent a fair amount of this time with the four reclaimers who participated in this study in an effort to build relationships with them that would enrich my experiences with them in the field and the information that they would share.

Bekezela is an abandoned building located under the M1 bridge on the corner of Goch and Carr Street in Newtown, Johannesburg. The building was once a college known as Bekezela College but was closed down eleven years ago (Sejake, 2016). Today, up to a hundred people call it home. Most of its inhabitants work as street reclaimers in Johannesburg. Those who do not collect materials from the streets live with people who do such as partners, family members or friends (Pather, 2017). Bekezela is a Zulu word which means endure or persevere. The word perfectly describes the essence of the people who reside there. Collecting waste and a heavy trolley with said waste across substantial distances requires endurance and perseverance.

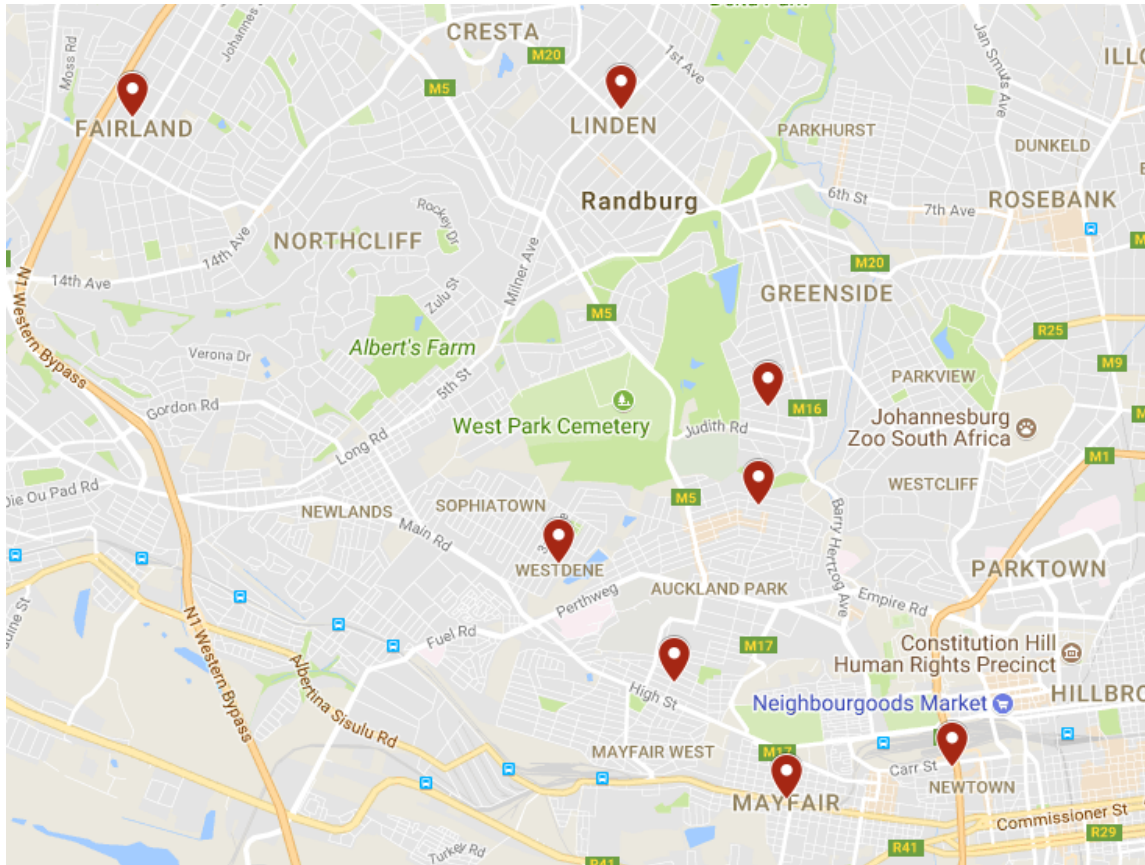


Figure 1. Study Area with place marks indicating the location of Linden, Fairland, Mayfair, Westdene, Brixton, Emmarentia and Bekezela (MyGoogleMaps, 2017)

I was prepared to work with reclaimers who lived anywhere around Johannesburg as long as they resided in the same area so that their movement and other spatial decisions that they make can be initiated from a common place. The movement of the different genders, for example, would be noteworthy if they all begin their journeys from a similar location. I was informed about Bekezela by an informant who was introduced to me by my supervisor. The informant advised me to find people who work in Bekezela because there are numerous street reclaimers who live there. Furthermore, Newtown is in close proximity to a number of suburbs that street reclaimers frequent. Thus, I could observe the reclaimers in various environments

3.2 Methodological Approach

Mugenda (1999), maintains that qualitative research is intended to uncover the participants' behavior and points of view that drive them with reference to specific topics or issues. The present study used a qualitative approach to determine how gender influences the physical mobility of four street reclaimers. The qualitative approach was the best approach to use because I needed to gain insight into the street reclaimers' behaviours and points of view that drive them to make certain spatial decisions related to their movement around the city. I used three qualitative approaches: phenomenology, ethnography and field research in this report.

Firstly, I used phenomenology which emphasizes a focus on people's subjective experiences and interpretations of the world (Merleau-Ponty and Smith, 1996). I interviewed street reclaimers in an effort to understand how their subjective experiences, perceptions and interpretations of the world influence their physical mobility in Johannesburg as street reclaimers.

Secondly, I utilized an ethnographic approach which generally involves observing a person or groups of people as they go about their daily activities and documenting the observations (Emerson *et al.*, 2011). This approach requires the researcher to actively take part in whatever daily activities the participants would be taking part in at the time (Emerson *et al.*, 2011). According to Steckler *et al.* (1992) ethnography helps to reveal aspects of the participant's life that are unarticulated. Furthermore, it helps the researcher to observe the participants behavior in different contexts of everyday life.

I believe that applying these two qualitative approaches was useful because they complement each other. On the one hand, the interviews revealed the elements that

affect the manner in which participants perceive space and their ability to move through space. On the other hand, the observations and assimilation into the lives of the participants highlighted issues that the four street reclaimers themselves may have been unaware or did not want to discuss.

Lastly, I conducted field research that involved going into the field to observe the participants and the spaces in which they move in their natural state or in situ (Mugenda, 1999). I travelled with each of the four participants (two men and two women) separately and obtained notes, pictures, and voice recordings as well GPS tracks of their movement through space while they collected material.

3.3 Methods

Semi-structured interviews, observations and mapping techniques were used to support the qualitative approaches in this study. Below is a table containing the research objectives and methods that were used to achieve the objectives.

Table 1. Objectives and methods used to achieve research objectives

Objectives	Methods (How the evidence will be collected)
	-Observations

1. Find out how male and female street reclaimers navigate the streets of Johannesburg (mode of transport).	-Semi-structured or unstructured interviews.
2. Observe and Map the routes that street reclaimers take.	- Record GPS tracks of street reclaimers -Semi structured interviews -Observations (written descriptions of routes taken by reclaimers, voice recording, video recording).
3. Uncover how they decide which routes to take and where to go.	- Semi-structured interviews - Observations (written descriptions) -
4. Investigate and analyze key elements that street reclaimers consider before they navigate through space.	-Semi-structures interviews
5. Discover the factors that negatively or positively affect the physical mobility of street reclaimers.	-Semi-structured interviews -Observations
6. Gain insight into how reclaimers think gender affects physical mobility.	-semi structured interviews

Sampling Techniques

A sampling technique is a method that divides the population into a number of parts for the purpose of participating in some sort of research. Sampling is used because it provides a smaller group of people for researchers to work with and it makes data collection cheaper, especially if one has limited resources (Ritchie, 2013). In view of the fact that I used a qualitative approach in this study, I had to use a sampling technique that would include specific people with particular experiences.

Denscombe (2014), states that mixed purposeful sampling is the best sampling to use. Mixed purposeful sampling is a mixture of two or more sampling techniques. The sampling techniques I used are chain sampling and judgement sampling. I used judgement sampling because it is used when the researcher wants to choose a sample group that represents the attributes required for the research. That is, two men and two women who work as street reclaimers in Johannesburg.

I was introduced to the participants by an informant who was introduced to me by my supervisor. The informant introduced me two men and one woman at a reclaimer meeting held in Ghandi Square. One of the participants later introduced me to another female reclaimer who became part of the four participants I worked with.

Semi-structured interviews

Semi-structured interviews are formal but they are not as rigid as structured interviews. The interviews include open-ended questions and an interview guide which is basically

a list of questions and topics that need to be covered during the interview (Bernard, 1988). The benefit of semi-structured interviews according to Bernard (1988), is that they are best used in instances when one will not get more than one chance to interview a participant. Semi-structured interviews were the best type of interviews to use because it would have been difficult to ask participants questions multiple times. Appointments had to be made multiple days before we met and regularly just after I interacted with another reclaimer. It was quite difficult to get a hold of them as most of them did not own phones.

I conducted the interviews at Bekezela with each reclaimer after I travelled with them and carried out observations. After losing my phone that contained voice recordings of the interview responses I had to conduct interviews in two consecutive days. I interviewed two reclaimers per day.

Observations

In an effort to understand the spatial decisions that reclaimers make I conducted participant observations and non-participant observations. Fetterman (1998), sustains that participant observations involve the researcher's participation in the lives of the people being studied with maintenance of a professional distance that allows adequate observation and recording of data. I accompanied individual reclaimers while they collected their materials for two. This will helped me to understand the factors that influence their movement better because their self-reporting (asking people what they do) was sometimes different from their actual behaviour (what people actually do). Atkinson and Coffey (2003), agree that participant observations have an advantage

over other data collection strategies in studies involving the day-to-day activities of people and their spatial decisions because the researcher is able to observe events and ask the participant the meaning of the event during and after the event has occurred.

I also conducted non-participant observations where I observed the spatial surroundings and examined the details of how reclaimers interacted with each other. I used a voice recorder, pictures and field notes to record the observations I made.

The participants allowed me to choose the days of the week that would be most convenient for me to conduct field research (participant and non-participant observations as well as recording GPS tracks of reclaimers). The only condition was that it could not be during the weekend (Saturday and Sunday) because they do not work on those days. The participants informed me that they use the time to relax and unwind after a week filled with exhausting work. The reclaimers go to a different place every day of the week. In that regard, I chose a day and they told me where they would be going that day.

I wanted to adequately compare the movement of men and women when I recorded GPS thus, I travelled with one female and one male reclaimer on two Mondays in two separate weeks:

Table 2. Travel roster with the days of the week travelled, street reclaimer accompanied and places that were travelled to.

Week	Day of the Week	Street Reclaimer	Place Travelled
Week 1	Monday	Male Reclaimer #1 (MR#1)	Linden
	Tuesday	Male Reclaimer #2 (MR#2)	Westdene

Week 2	Monday	Female Reclaimer #1 (FR#1)	Brixton
	Tuesday	Female Reclaimer #2 (FR#2)	Mayfair
Week 3	Thursday	Male Reclaimer #1 (MR#1)	Fairlands
	Friday	Male Reclaimer #2 MR#2)	Emmarentia
Week 4	Thursday	Female Reclaimer #1 (FR#1)	Fairlands
	Friday	Female Reclaimer #2 (FR#2)	Linden

During the observations, I pulled a trolley and collected waste for the reclaimers I was working with once for each reclaimer (four in total). I would of ten travel with a friend or colleague to help me pull the trolley when I got tired. The other person wuld also take pictures if I was pulling a trolley. I also recorded GPS tracks, took notes, recorded conversations that I had with the reclaimer I was travelling with and took pictures when we stopped to rest on the way to the collection site and from Bekezela..

Mapping

Participatory mapping is a process of map making by non-experts who want to illustrate the relationship between land, ordinary people and their everyday experiences. The maps can be developed on complicated geographic information systems (GIS) or on paper. Mapping done on paper surfaces with participants helps to lessen the intimidation that those being investigated may feel. This allows participants to identify and define the issues and experiences that are important to them because they can add or detract from maps easily (Chambers, 2006).

One of this report's objectives was to map the gendered movement of waste pickers. Each reclaimer was given an A4 page and stationary to draw the route that they took on

the day that I travelled with them. Upon arrival in Bekezela from collecting materials from either Linden, Fairland, Brixton, Emmarentia, Westdene or Mayfair (see *table 2*), the reclaimers would store their materials in a communal storing area where most reclaimers would keep their materials until they sold them at the end of the week. The reclaimers would then proceed to draw their routes from memory. It was important for the reclaimers to draw the maps soon after they had just come from collecting because the routes were still fresh in their minds.

The GPS tracks that I used in the maps I produced were obtained on the days that I conducted my participant and non-participant observations. I used a Garmin eTrex 20x GPS to record the street reclaimer's tracks. The places that the street reclaimers travel to each day to collect material depends on Pikitup's waste collection schedule. Pikitup is the official service provider of waste management in the City of Johannesburg. (Muzenda, 2013). Pikitup collects waste on different days in different neighbourhoods. On Monday, for example, Pikitup collects waste in Brixton and Linden. The reclaimers are aware of the schedule and travel to those places. Street Reclaimers wait until residents place their bins in the streets for Pikitup to empty.

Developing the map involved importing the GPS tracks I obtained from the GPS into www.mygooglemaps.com where the tracks were illustrated. The maps were further edited using Microsoft paint and word. Places of interest that the reclaimers drew on their maps and spoke about during observations or in interviews were included on the map and represented by particular symbols. The areas where reclaimers mentioned that they always feel unsafe when they pass by were also indicated. The maps visually

illustrate the routes that male and female street reclaimers used, places of interest and areas where they felt unsafe.

Limitations

The most noteworthy limitation I encountered was safety. Bekezela is considered to be the most dangerous place in Newtown. Most of the people who reside in Bekezela are poor, uneducated and regarded as criminals by authorities (Sejake, 2016). I had performed a bit of research about Bekezela and was aware of its image as a dangerous informal settlement situated in the city. The first time I went to Bekezela, the informant who introduced me to the participants in this study warned me to never walk in alone or with another person for that matter without being escorted by someone who lives in Bekezela. I communicated with him regularly and would ask him to pass on messages to the participants because they did not have cellphones. During the duration of my fieldwork he kept reiterating that I should never walk in without being escorted by someone who lives in Bekezela because I will likely be mugged or victimized in some way.

The reclaimers leave very early when they travel to their collection sites so I had to be outside Bekezela, waiting for them to appear in the early hours of the morning around 3:30 am and 4am. The streets were dark and almost deserted until the reclaimers walked out of Bekezela. I never really felt safe when I was on my own but when I walked in with one of the participants everyone inside would greet me with smiles and would ask me questions. Having to wait for someone to fetch me from the outside was at times tedious because the informant would sometimes not answer his phone and I

would be waiting outside for hours on end. At some point I asked one of the residents to give their phone number so they can ask one of the participants to fetch me from the outside.

After a few weeks of waiting outside for an escort, one security guards that patrol just outside Newtown asked me why I am always standing outside such a dangerous place. He wanted to know what I did inside and how anyone would actually want to go in there. I informed him about my research, the participants and the fact that I have to wait outside for someone to escort me in. He sympathised with me and asked one of the other security guards to walk in with us. He then said I should ask one of the security guards to walk me inside if no one answers their phone again. I did just that and they were all willing to do so.

Another limitation was losing one of the initial participants I had because he moved away from Bekezela before I could interview him and complete one more day of observations with him. I quickly found another participant with the help of the informant.

I presented participants with a participant information sheet and asked them to sign two consent forms that stated that they agree to be interviewed and allow me to travel with them and record their movement respectively. In both consent forms it states that I can include anything they say during the interview sessions and discussions we had while we navigated the streets of the city. The participants were given consent forms that were in the language of their choice. All of them chose isiZulu. I also verbally explained the contents of the consent forms and made sure they understood everything they were consenting to. I also made sure that the participants are older than eighteen.

In conclusion, the semi-structured interviews helped me to get all the information that that I needed and left room for other questions that arose from responses that reclaimers gave. The interview process was not as rigid as it would have been with structured interviews. This allowed me to have a conversation with reclaimers and made them more relaxed. The participatory maps gave participants the freedom to show their interpretations of the space around them and the features of the built environment that are important to them. The observations helped me to understand the places that the street reclaimers spoke about during interviews and cultivate a deeper understanding of their lived experiences as street reclaimers. There were some limitations but I managed to get help and achieve the objectives set out in this report due to the methods that I used.

4. Results and Discussion

This chapter will discuss the results that were obtained from field observations, semi-structured interviews, participatory mapping and the recording of the reclaimers' GPS tracks.

4.1 Walking: the poor man's lot.

Most reclaimers use walking as their mode of transport when they collect their materials from various households and businesses. This mode of transport is the cheapest way for them to move in and around neighbourhoods with their trolleys which put their reclaimed material in. Though it is the cheapest mode of transport it is not the easiest and most effective form of transport to use for a reclaimer. I interviewed two male and two female reclaimers who each complained about having no choice but to walk because in their world that is the only option they have. When asked what mode of transport she uses to get to the areas where she collects her materials, female Reclaimer #2 replied:

"I use my legs. I walk everywhere I need to but I hate it. You know, I wish I had money so I can hire transport to take me to my streets and bring me back. Maybe one day. But now I can't because I barely make enough to survive. Transport is something you think about when your stomach is full."

The reality of this statement rings true for numerous poor South Africans. A qualitative study by Venter *et al.* (2013) revealed that high income participants prioritised efficiency (in terms of time, direct routes) with regards to the mode of transport they use. They agreed that they would rather pay more money to maintain their private vehicles than

take public transport or walk *any* distance to get to their destination or public transport. Middle income participants prioritised affordability as well as time but they agreed that they would spend more money if they need to get to their destination faster. Low income participants were concerned with having access to transport and affordability. They agreed that they would rather walk to their destinations than take public transport because if they do they will end up working for transport, leaving no money for other needs (Venter *et al.*, 2013)

The two male reclaimers agreed that only having walking as an option limits their mobility because they cannot go to places like Rooderpoort as much as they would like because it is very far from Bekezela. They said that they would have to leave for Rooderpoort the day before Pikitup is due to collect waste because this is when residents take out their dustbins to be emptied. They leave in the afternoon and arrive at night at their designated streets where they camp out and wait for residents to take their dustbins out the next morning.

During my observations I experienced the difficulty of walking in the streets, hauling a sack full of materials while trying to manoeuvre through oncoming traffic. Sometimes the sacks fall in the street and motorists express their displeasure or motorists refuse to give reclaimers right of way. Taxi drivers were the biggest problem

“Taxi drivers make it difficult to walk. They want us to use the pavement but the pavement is too small and the trolley doesn’t fit. I am a woman, I will be carrying a heavy load and the taxi driver will be rushing me to pass. One time a taxi driver threatened to bump into me. I was very scared because my little boy was sitting inside the sack on the trolley.”

- (Female Recycler #1)

During my field observations I also noticed that reclaimers do receive a fair amount of hostility from the motorists but there are some friendly people who hooted and said hello when we passed by or gave us time and space to cross the road.

“Before people used to complain when they saw us walking in the roads but now they are used to us. Some of them even greet or stop to give us food”

- (Female Reclaimer #2)

Considering that walking does take a very long time and is often times tedious, the reclaimers have adapted and found a way to transform their trolleys into a form of transport. They ride their trolleys for fun and speed. Picture a person racing down a steep snowy slope on a snowboard. That is more or less what it looks like when they ride down the streets. Some reclaimers even view their trolleys as cars and most of them are named after luxury brands.

“I call it Gusheshe (a slang word given to the BMW 325 iS model which means ‘panty dropper’). I drive it like a car. Did you see how fast it is!?”

- (Male Reclaimer #1)

While the city is asleep at 3 am in the morning, the reclaimers can be found driving their luxury cars that consist of two plastic squares held together by a piece of string. I had the privilege of being given a lift on these trolleys numerous times and to say it was a rush would be an understatement. I was not aware that I could feel dizzying exhilaration and crippling fear at the same time. The only thing that separated me from the gravelled road was two shaky plastic squares and a prayer. The first time was with Male Reclaimer #1. He used his body as a steering wheel and his feet as the brakes. At the end of the trip I had tears in my eyes. I am still not certain if they were from fear or

the wind. After riding a few times on the trolley I began to relax and take in my surroundings. We were barrelling down Annet Street when I finally understood why some people refer to Johannesburg as 'Jozi maboneng (Johannesburg the place of lights)'. Travelling at that speed with no separation between myself and my surroundings, the city looked like it was ablaze with light.



Figure 2. 4 am on Annet Street: Street Reclaimers waiting at the bottom of a hill for the other reclaimers to catch up.

Reclaimers adapted to their surroundings by riding down hills to save time and to get where they need to earlier in an effort to avoid traffic because it is more difficult to walk with or ride a trolley while there are cars on the road. Only one of the reclaimers that I travelled with did not ride her trolley because she considered it too dangerous. Riding the trolley does make it easier for reclaimers to get to the places that they need to but it is no substitute for a truck or car. The weight of the material they carry, weather conditions, traffic and the physical landscape of the city make movement slow and arduous.

It is clear that money gives one the flexibility to choose the most convenient and efficient way to travel. The participants are restricted from going to certain places

because it is too far. The trolleys that they use are also bought at a price. Those who can afford them are able to ride them and arrive at their destinations faster. Conversely, even with trolleys they cannot access some areas due to the lack of money to hire transport. This limits their ability to collect the valuable materials that they obtain from suburbs that are further away from Bekezela.



Figure 3. Reclaimer and researcher pulling trolleys through the streets of Westdene

I conducted eight days of observations where I walked with reclaimers from Bekezela to Westdene, Fairland, Linden, Brixton, Emmerentia or Mayfair. In those eight days I did not see any female reclaimers walking in the streets. The only female reclaimers I saw were the two I walked with. The absence of female reclaimers reiterates the argument that streets are dominated by men as Chenal (2014), stated in his book: *Urban space and models of urban planning*. He discovered that only 10% of the population in the streets of Nouakchott, Mauritania were women. The presence of women in the streets was due to sales or the market. This finding relates to the results from a study that Almahmood *et al* (2017) carried out. They found that there are a limited number of women in the streets of Saudi Arabia. The women who are seen walking do so to go

the mall or other shopping centres. I asked female reclaimer #2 why there were no other female reclaimers in the streets? She replied:

“Before 2010, there were lots of women who did this job but towards 2011 a lot of them stopped because many men started collecting materials at that time. We did not have designated roads in those days like we do now. Everyone worked together and went wherever the other person did not go. The Sotho men and foreigners came and did not want to share material. The women did not want to fight.”

There are various reasons for the lack of female reclaimers in the streets. Though there are women who collect materials, we did not get to see them because they would leave their homes later in the morning and arrive later in the afternoon once we had already come back. These women do not reside in Bekezela. They live in areas such as Fietas, Booyens and Doornfontein. There are only two women who still collect recyclables in Bekezela.

4.2 Threat of crime and Experienced Crime

Reclaimers leave very early in the morning to collect their materials. On the day that Pikitup is due to collect waste, the reclaimers make sure that they collect any value materials before 11am when Pikitup trucks show up. On average, they would leave Bekezela at 5am if they are going somewhere close like Mayfair, Westdene, Linden or Melville. On Thursdays the men would leave at 3:30am or 4am to go to Fairlands. At that time in the morning there is a minimal number of cars on the roads and it is still dark. Due to that reason the two female participants often travel with a man. Female reclaimer #1 travelled with her boyfriend while female reclaimer #2 would travel with a male friend when he was not collecting materials at places further than she was going. On days that her friend wanted to travel further like on Thursdays when most of the

men go to Fairlands, she sometimes went with him but he did not always allow her to do so.

“He says I walk too slowly and I take too long. He doesn’t want me to go with him because he says I waste time and I don’t have the strength to keep up”

- (Female Reclaimer #2)

The female reclaimers revealed that they walk with men because they do not feel safe enough to walk on their own, even in the afternoon. They rarely walk around with their trolleys on their own but when they do they only do so when it is not dark outside.

“I don’t like walking alone without another man with me because I am afraid my trolley can get stolen. When I walk alone I wait until the sun comes out before I go. I make sure that I dress like a man so people don’t see that I am a woman and try to take advantage.”

- (Female Reclaimer #1)

Female Reclaimer #2 had similar fears:

“I don’t leave very early on my own. Because I am a woman I worry about my safety. I may be hit by a car, raped or have my belongings taken.”

The female reclaimers seemed to be gravely concerned about my safety when I travelled with them even though we were accompanied by a man. I felt a sense of anxiety that I did not feel while I was walking with the two male reclaimers. I suddenly thought of all the things that could happen to me while we were walking through the streets. I could be kidnapped, raped or someone could just run over and grab my backpack. I found myself holding on to a can of pepper spray I kept in my pockets in

case something happened. While on our way to Linden one Friday, I asked Female Reclaimer #2 if anything had ever happened to her while she was walking in the streets. She said that she had been mugged once back in 2006. She had been walking alone in Emmerentia when two men asked her for her trolley but she refused and told them that she needed it for work. They used force and took the trolley and some of her belongings. She tried to defend herself but they overpowered her.

I was impressed that she put up a fight against two men who were stronger than her. She informed me that at the time she had just been released from prison after serving two years of her four year sentence for stealing copper cables. Her time in prison was filled with lots of fights and some bloodshed so she had gotten used to defending herself.

“You have to transform yourself into someone you are not in prison to survive. I don’t like fighting but in prison I had to learn how to defend myself. If you don’t change you won’t make it. It’s like in the streets. When you are a woman on your own and do this work you must transform yourself and be a man. You can’t wear dresses and look weak.”

Following the incident with the two men, Female Reclaimer #2 started walking with a man because she did not want to be mugged again. Though it has been eleven years since she experienced any form of victimization, she still maintains that the streets are unsafe.

Female Reclaimer #1 admitted that she had never experienced any sort of crime while she was walking around but she still feels unsafe. The fear of crime keeps her from walking alone in the streets to collect materials. She informed me that she has been walking with her partner since 2012 because she felt like Bekezela and the surrounding

areas had become unsafe. The reasoning behind her deduction is that she heard of female reclaimers being victimised by men from Lesotho, Mozambique and Zimbabwe.

The insecurity that the two women experience in public space is a reality. Conversely, the two men were comfortable in public space. I experienced a fraction of the freedom that they enjoy in urban streets on a chilly Thursday morning while Male Reclaimer #1 and I walked to Fairlands. There were other male reclaimers who would occasionally pass by and greet us but for a majority of the trip it was just the two of us. We used the same roads that the female reclaimers use when they go to Britton, Mayfair or Melville. I still had remnants of the fear that I had experienced the previous two weeks that I had been walking with the two female reclaimers. I made this known to Male Reclaimer #1 and he assured me that I would be safe because I was walking with him and not the female reclaimers. I was stunned by the statement for a few minutes but after a while I realised that I did not feel the same level of anxiety that I did the previous two weeks. It felt like Male Reclaimer #1 was transferring his confidence to me. The certainty that he felt about his safety made me feel safer. After an hour of walking and being transported on the trolley, I began to enjoy the fact that there were no people in the streets instead of being afraid that if something happened, no one would heed my cry for help. The other male reclaimers were riding down the streets with us and it felt like we owned the streets.

The insecurity concerning my safety that I experienced while I was travelling with the two female reclaimers was based in part by the uncertainty that they felt about their own security and their concern about my safety. Fear based on perceived threats to safety is just as arresting as fear gained from experienced victimization.

4.3 Participatory Mapping

The reclaimers were each asked to draw a participatory map of the routes they use. Most of them were only familiar with the names of the routes that they took and did not know much about the surrounding areas two to three blocks from their designated streets. When asked how they choose the streets they use, most of the reclaimers stated that other reclaimers who had already been reclaiming for a while taught them which streets to use. I asked them why they did not walk around and find the best streets to use themselves. Two of the male reclaimers revealed that they do not like walking around rich neighbourhoods because they are regarded with hostility and seen as thieves.

“They think because I am black and a man I am a criminal”

- (Male Reclaimer #1)



Figure 4. Male Reclaimer #2 Drawing his route to Emmarentia.

The maps that the reclaimers drew and the GPS tracks aided in creating maps that reveal various details. Each map illustrates the GPS track of one female reclaimer and one male reclaimer in a day.

Map 1: MONDAY

Movement of MR#1 and FR#1 on Monday

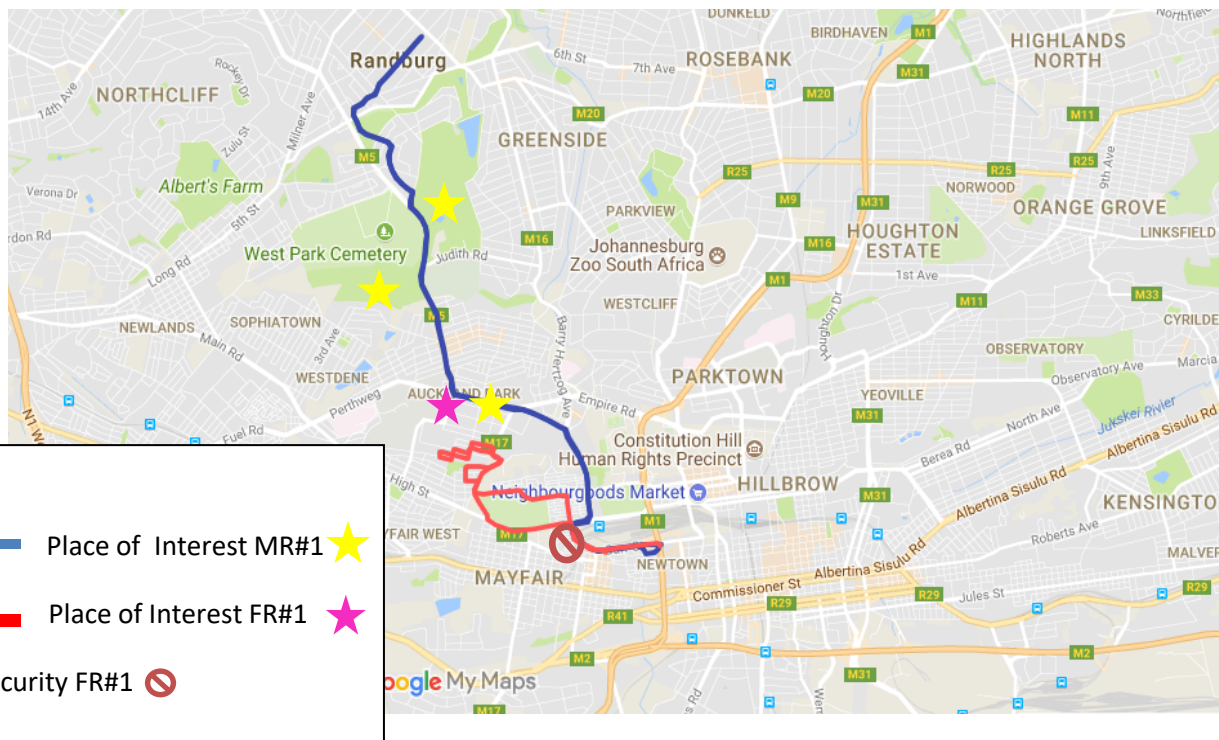


Figure 5. Typical movement of Male Reclaimer #1 to and from Linden and Female Reclaimer #1 to and from Brixton on a Monday.

Male Reclaimer #1 and I walked for 21 kilometres to and from Linden in the first week of my fieldwork. We left Bekezela quite early at around 4am so we could get to Linden around 5:30 am to 6 am. Male Reclaimer #1 did not show any concerns for his safety or mine when travelled together. He also did not show any signs of anxiety or discomfort. In that regard, there are no symbols indicating areas where he felt unsafe. The places of interest that MR#1 included in his map and spoke about are the West Part Cemetery, Annet Road and the Johannesburg Botanical Gardens. He mentioned that he would like to be buried in a cemetery as nice as the West park cemetery one day and not in Lesotho because the cemeteries there are not as beautiful.

“If I can’t sleep somewhere nice when I am alive at least I can sleep forever in a nice place like this. I like this place because in there, everyone is the same. No one is better than anyone else.”

- (Male Reclaimer #1)

MR#1 also mentioned that he enjoys walking past the Johannesburg Botanical Garden because he likes parks and would like to take his daughter there one day. His daughter lives in Lestho with her mother and both mother and daughter are not aware of what MR#1 does for a living. He revealed that he just sends money home and tells them that he works as a mechanic. I asked him why he felt the need to lie to his family. He confessed that he thinks they would not respect him if he did. The last place of interest is Annet Street where the reclaimers would race down the steep slope towards the Reya Vaya Station. On both occasions that I travelled with him he would urge me to walk faster so he could catch up to the other reclaimers so that they could race down the slope.

Female Reclaimer #1 and I travelled 9.18 kilometers to and from Brixton. We left round 6 am and arrived at 7 am. She was relaxed but anxious about passing under the bridge on Carr street. Though there were cars cars passing through, she said she always feels nervous when she passes through even in the afternoon because people often sleep under it on the sides of the road or above it on the pavements. She she heard of a man being mugged and having their trolley taken. A point of interest that FR#1 drew and mentioned during observations was the University of Johannesburg. She mentioned that she passes by the university regularly and always wishes her eldest daughter who is 20 years old would attend there.

“You know every time I pass here I wish I could see my daughter inside there. I don’t want her to be like me. I want her to be in there, not here [in the streets].”

- Female Reclaimer #1

Movement of MR#2 and FR#2 on Tuesday

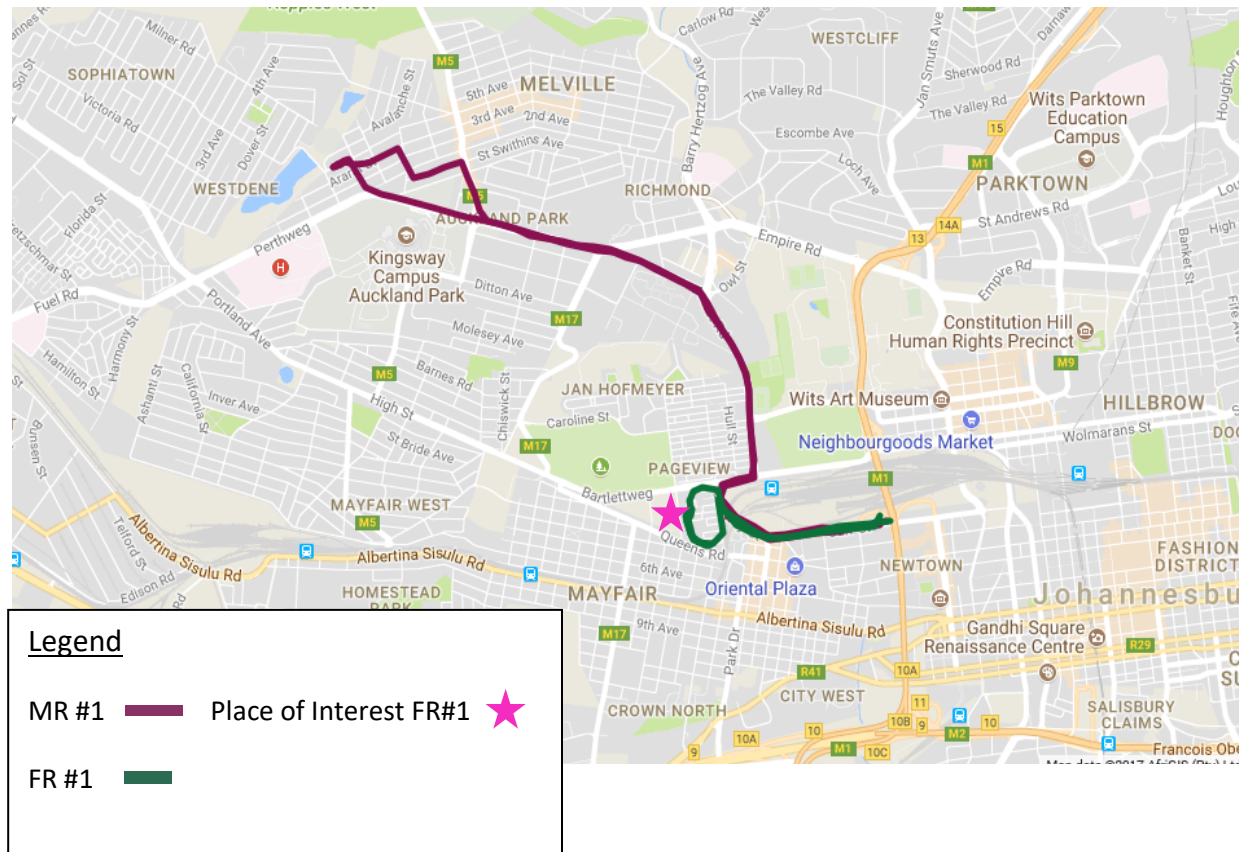


Figure 6. Typical movement of Male Reclaimer #2 to and from Westdene and Female Reclaimer #2 to and from Mayfair on a Tuesday.

Male Reclaimer #2 and I travelled 11.6 kilometres to and from Westdene we left at 4:30 am and arrived at . Female Reclaimer #2 and I travelled 4.12 Kilometres to and from Mayfair we left at 6:30. The two reclaimers did not show or express any concerns about safety. However, there was a mild disagreement between FR#2 and a male street reclaimer who took materials out of the dustbin on her designated road. The male reclaimer had knowledge of the fact that the road is designated to her but he took it anyway. He was under the impression that FR#1 did not have her partner with her that day so he took advantage. Her partner was collecting materials for her from another

steet so we could finish collecting faster. The partner was called and he spoke to the male reclamer.

Only FR#1 had a place of interest worth mentioning because we spoke at length about religion and its connection to race. There is a mosque in Mayfair that RF#1 mentioned she would like to see on the inside but was afraid to go because she had never seen any black people go into. I explained that Mosques are essentially churches for muslims. After that we had a long discussion about how race is connected to religion. She mentioned that she does not believe in Christianity anymore because of all the strange things that she has seen happen in churches, like people eating snakes and drinking petrol. She was also very critical of black people in general because according to her only black attend churches that behave like that. Furthermore, her idea of Christianity is rooted in black people performing and witnessing weird acts. She mentioned that she will never become a Christian because it gets one nowhere.

“Black people are Christian but they’re the poorest. They just goto church and cry and faint then they go back home and do nothing.”

- (Female Reclamer #2).

Map 3: THURSDAY

Movement of MR #1 and FR#1 on a Thursday

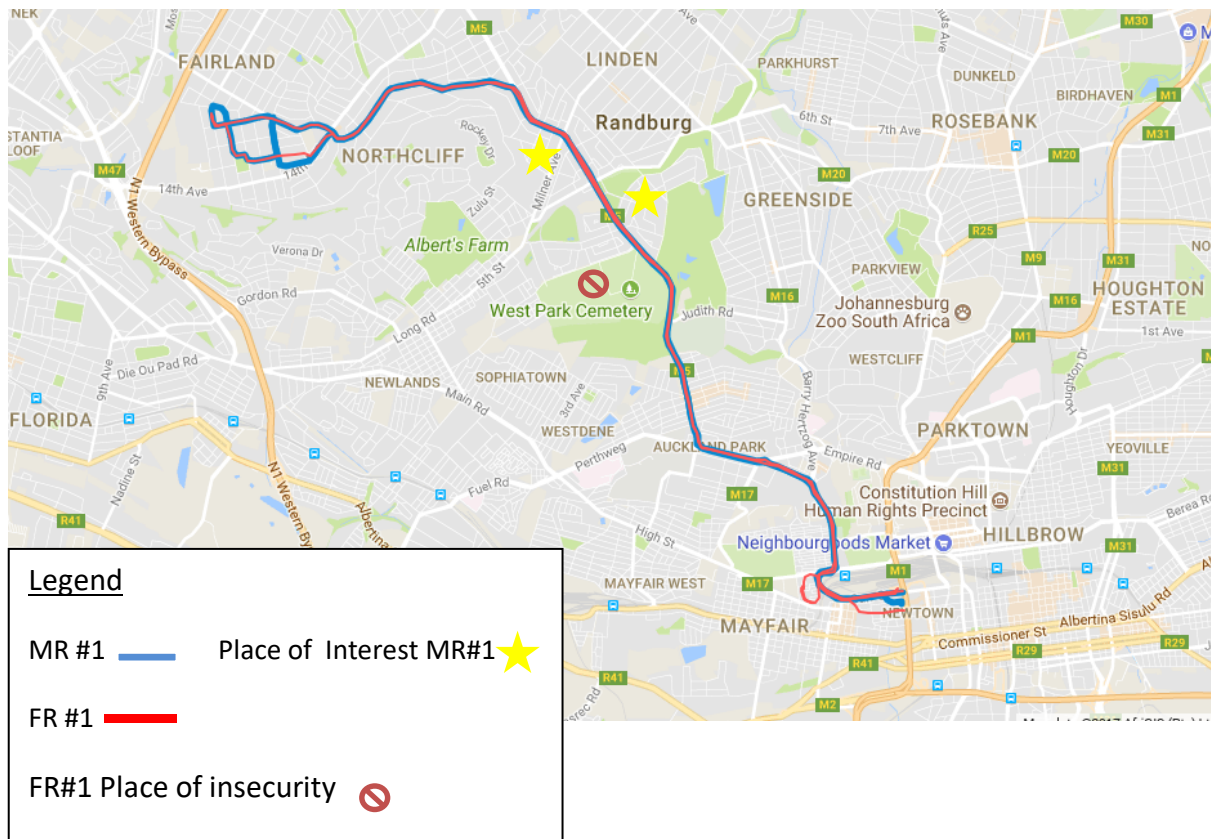


Figure 7. Typical movement of Male Reclaimer #1 to and from Fairland and Female Reclaimer #1 to and from Fairland on a Thursday.

I travelled 39.2 Kilometers to Fairlands with Male Reclaimer #1 we left at 3:30 and arrived at 6:30 am. Places of interest on Thursdays MR#1 are the area outside the Botanical Garden and before 6th street and a business along the M5. A large number of reclaimers travel to Fairland on Thursdays so on the way to and back from Fairland, a lot of them rest on a patch of grass outside the Botanical Gardens and share cigarette or homemade beer or other store bought alcohol that they carry in bottles. Some of them bring more than one bottle to share. According to MR#1 it has become a ritual that he looks forward to on Thursdays. MR#1's other place of interest is a business along the M5 that saves boxes and bottles for him to collect on Thursdays. I travelled 31.2 kilometers to Fairland with Female Reclaimer #1. She had no particular points of interest but she expressed fear when we passed by the cemetery. She strongly believes that there are spirits

lingering in and around the cemetery. We left Bekezela at 3 am. By the time we passed the cemetery it was 4 am and still dark. The streets are quite ominous at that time. She showed visible distress and discomfort. Her partner and I had to reassure her that she will be okay. FR#1 did mention that she does not enjoy walking to Fairland because it was too far and it was always a struggle to pull the trolley because her partner always insisted that if she goes with him she must be willing to pull her weight. She revealed that she only agreed to go to Fairland that day because I was there to help her with the trolley and she had cooked and cleaned the night before.

Movement of MR#2 and FR#2 on Friday

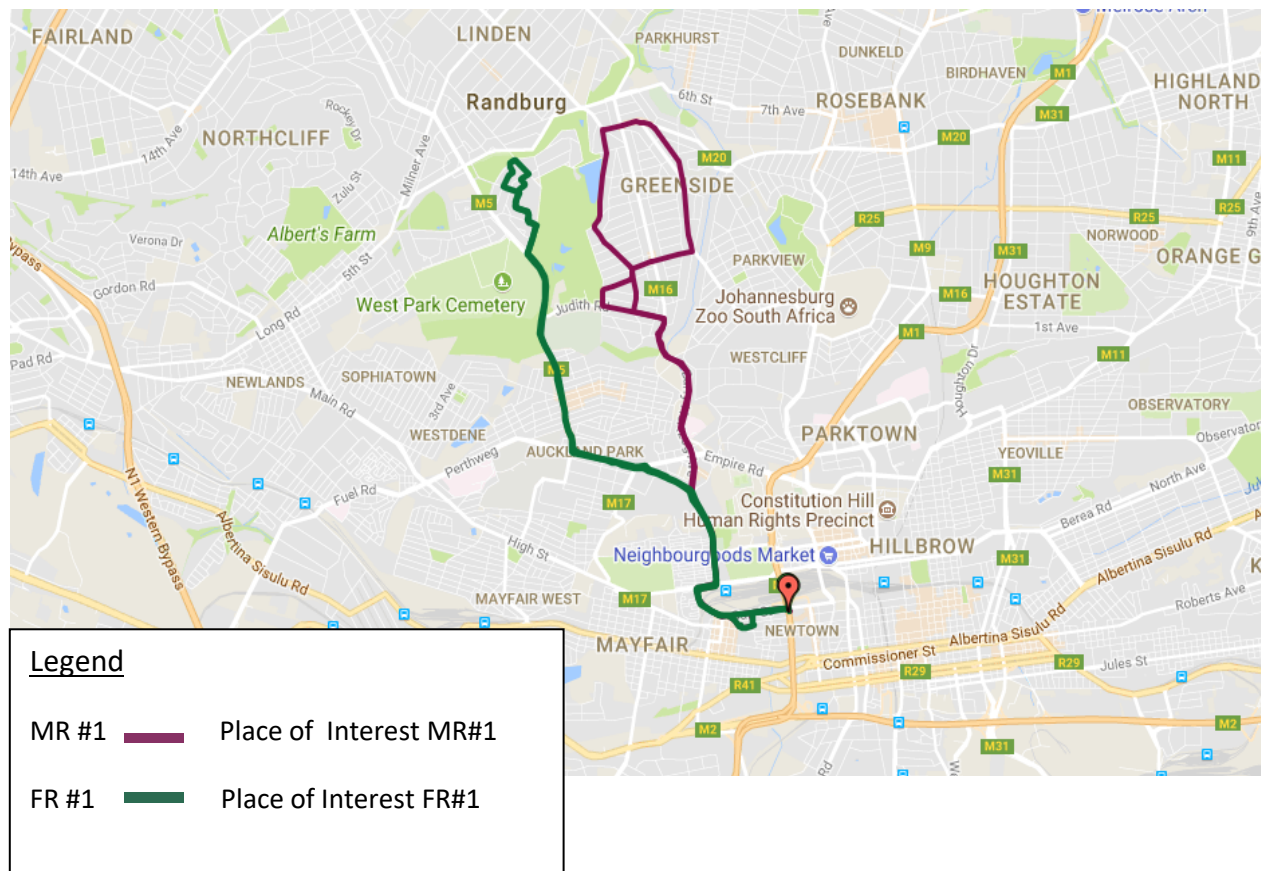


Figure 8. Typical movement of Male Reclaimer #2 to and from Emmarentia and Female Reclaimer #2 to and from Linden on a Friday.

Travelled 18.3 kilometers with Male Reclaimer #2 to and from Emmerentia and 17.8 Kilometers with Female Reclaimers #2 to and from Linden. On these days the reclaimers did not express any feeling of insecurity. On both occasions we left Bekezela at 6 am.

It was evident from the movement of the reclaimers that the female reclaimers travelled a shorter time compared to the male reclaimers. From the interviews, observations and discussions I had with the reclaimer, this can be attributed to the fact that the reclaimers prefer to go to places that are not too far from Bekezela because they have different

chores to do or take care of their children. Furthermore, they find it difficult to carry the trolleys because they are too heavy.

4.4 Gender Roles and stereotypes

Female Reclaimer #1 has three children. She revealed that she is unable to go to places that are further away from Bekezela such as Fairlands, Rosebank and Rooderpoort because she has to be at home to take care of her children. Her partner also disagrees with the work she does and calls it 'dirty' and a 'disgrace' because she spends so much time in the streets. Due to her responsibilities as a mother and her partner's opinion about her job, she tries to limit the amount of times she goes out to collect materials. She has also tried to look for other work such as working in a shop because she would not have to walk around as much.

4.5 Physical Aspects of the Built Environment

It has been proven that certain aspects of the built environment have the potential to limit or promote physical mobility in the urban city (Wang *et al.*, 2016; Rocheleau *et al.*, 1995; Almahmood *et al.*, 2017). The four reclaimers that I interviewed stated that they choose the routes they use based on the availability of bus lanes so that they can pull their trolleys in an area that does not have cars. Furthermore, they use roads that are wide and open as opposed to narrow roads because it is easier for car to overtake them. The topography of the landscape such as elevation is not a hindrance. They stated that they would rather walk up a very steep hill if that was the shortest route than use a route with a more uniform elevation that was longer. Their first priority is arriving at their destination in the shortest possible time.

5. CONCLUSION

Physical mobility of waste reclaimers demands that they navigate the different spaces where waste is found, particularly in an urban area such as the City of Johannesburg. This means that the furthest they are able to move around and collect waste in different areas is how they are able to benefit by making more money. This study has proven that mobility and space are gendered because women and men experience space differently. Furthermore, they make different spatial decisions with regards to their movement based on their narratives.

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Appendices

Interview Schedule



Interview Schedule for Street Reclaimers

My name is Amogelang Mokobane; I am a Geography Honours student. As part of my studies at the University of the Witwatersrand, I am undertaking a research project entitled: Mapping the Gendered Movement of Street Reclaimers in Johannesburg. This research will provide insight into the various factors that influence the movement of male and female street reclaimers. I would like to invite you to take part in an interview that will be done in English, seTswana, Sepedi or isiZulu depending on the language you prefer. The interview will be recorded using a tape recorder, but only with your explicit consent. I may also take pictures of you but this will also only be done with your permission. The interview may take 30-45 minutes. I have a consent form for you to sign if you are willing to participate in this interview. Any information that you provide will be treated with confidentiality and anonymity is guaranteed. I require information from you to answer the following questions:

Participants	Schedule
Street Reclaimers	1. Mode of transport used. 2. How long it takes to travel. 3. Routes used to travel from collection site, to sorting site to buy-back centres. 4. Factors that influence which routes reclaimers use. 5. Major challenges faced when navigating the streets. 6. How challenges restrict the movement of reclaimers. 7. Factors that male reclaimers think restrict the movement of female reclaimers and the routes they use. 8. Factors that female reclaimers think restrict the movement of male reclaimers and the routes they use. 9. Challenges unique to female reclaimers that men don't experience. 10. Challenges unique to male reclaimers that women don't experience. 9. Does gender influence the routes that street reclaimers use?

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Observation Guide



Observation Guide for Street Reclaimers

Observation of the Movement of Street Reclaimers through Johannesburg

My name is Amogelang Mokobane; I am a Geography Honours student. As part of my studies at the University of the Witwatersrand, I am undertaking a research project entitled: **Mapping the Gendered Movement of Street Reclaimers in Johannesburg**. This research will provide insight into the various factors that influence the movement of male and female street reclaimers. I would like to travel with you through the streets while you collect, sort transport and sell your materials for a maximum of three days to observe the routes you take and to track your movements using a GPS tracker. Furthermore, I will ask you a series of questions while we move through the city. The observations I make while I travel alongside you will only be used for research purposes. I may take pictures and voice recordings of our conversations while we travel but this will only be done with your permission. I have a consent form for you to sign if you are willing to allow me to accompany you and track our coordinates while we travel. Any information that you provide will be treated with confidentiality and anonymity is guaranteed.

Observation Guide

Sub-Questions	Questions to Ask Self when Observing
1. How do male and female street reclaimers navigate the streets of Johannesburg? (mode of transport used)	1. What did we use to travel from the collection site to the sorting site? 2. What did we use to travel from the sorting site to the buy-back centre? 2. How long did it take?
2. What routes do street reclaimers use?	1. Which routes did we take?

	2. Did we use busy main roads or small roads and alleyways?
3. What are the key factors that street reclaimers consider when they choose their routes?	1. Are the routes near public facilities such as toilets, taps and places to buy food? 2. What are some of the reclaimers' comments about the surroundings? 3. Consider the infrastructure and elevation. Do these affect the reclaimers' movement?
4. What are the major challenges that street reclaimers face when they navigate the streets of Johannesburg?	1. Consider the comments that the reclaimer makes about the surroundings and people. 2. How does the reclaimer interact with other reclaimers and what do they talk about? 3. Do you feel any immediate danger or a sense of threat in certain areas? 4. How are people reacting to your presence? 5. How does the public interact with reclaimers in different areas of the city?
5. How do the challenges restrict the movement of street reclaimers?	1. Ask the reclaimer why you could not use another route instead of the one you're using. 2. Are there any visible physical aspects of the landscape that force the reclaimer to use other routes? 3. Do the comments that the reclaimer makes about the surroundings indicate that there are factors restricting their movement. 4. How far have you travelled? 5. How much material is the reclaimer carrying ?
6. How do reclaimers think gender affects their navigation to the city?	1. Are the male reclaimers taking routes that female reclaimers do not take? 2. Are there any visible factors that you can see that may be causing the difference in movement? 3. What are the relationship dynamics between male and female reclaimers?

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Verbal Consent form for Interviews



VERBAL INTERVIEW CONSENT FORM FOR STREET RECLAIMERS

Hello my name is Amogelang Mokobane, I am a Geography Honours student at Wits University. I am doing a research project that is investigating the factors that influence the movement of male and female street reclaimers.

As part of this project I would like to invite you to take part in an interview that can be conducted in English, seTswana, Sepedi or isiZulu, depending on the language that you would like to use. The discussion will be recorded using a tape recorder, but only with your explicit consent. I may also take pictures of you but this will also only be done with your permission. The discussion may take between 30-45 minutes.

Confidentiality and anonymity are ensured.

Please answer yes or no to the following questions to confirm your consent.

1. Do you agree to participate in this research project, which has been fully explained to you by me?
2. Would you like your participation in this interview to remain anonymous?
3. Do you agree that I may use anonymous quotes in this research report?
4. Do you agree that the interview may be audio recorded and you may have your picture taken?

..... (Signature)

..... (Name of participant)

..... (Date)

I, Amogelang Mokobane confirm that the above participant has been fully informed about the nature of the research.

Researcher: Amogelang Mokobane

Verbal Consent for Observations



VERBAL OBSERVATION CONSENT FORM FOR STREET RECLAIMERS

Hello my name is Amogelang Mokobane, I am a Geography Honours student at Wits University. I am doing a research project that is investigating the factors that influence the movement of male and female street reclaimers.

I would like to travel with you through the streets while you collect, sort transport and sell your materials for a maximum of three days to observe the routes you take and to track your movements using a GPS tracker. Furthermore, I will ask you a series of questions while we move through the city. The observations I make while I travel alongside you will only be used for research purposes. I may take pictures and voice recordings of our conversations while we travel but this will only be done with your permission. I have a consent form for you to sign if you are willing to allow me to accompany you and track our coordinates while we travel. Any information that you provide will be treated with confidentiality and anonymity is guaranteed.

Confidentiality and anonymity are ensured.

Please answer yes or no to the following questions to confirm your consent.

1. Do you agree to participate in this research project, which has been fully explained to you by me?
2. Do you agree to allow me to travel with you while you collect, sort and sell your materials?
3. Would you like your participation in this observation to remain anonymous?
4. Do you agree that I may use anonymous quotes in this research report?
5. Do you agree that the observation may be audio recorded and you may have your picture taken?
6. Do you agree to allow me to record your movements around Johannesburg with A GPS tracker?

..... (Signature)

..... (Name of participant)

..... (Date)

I, Amogelang Mokobane confirm that the above participant has been fully informed about the nature of the research.

Researcher: Amogelang Mokobane

Signature -----Date-----

Signature -----Date-----